Informe del método Simplex

Yo

2026

1. History of the Simplex Method

- George B. Dantzig developed the Simplex method in 1947 to solve linear programming problems.
- The method seeks an optimal solution among the basic feasible solutions, associated with the vertices of the feasible region.
- In each iteration, a variable enters the base and another leaves, resulting in a new basic feasible solution. This is known as pivot operation.
- In bounded problems, iterations advance toward an improvement of the objective; in degenerate cases, the value may remain the same.
- It can detect optimality, unboundedness, and, according to the formulation, infeasibility.
- It is efficient in many practical problems, although its worst-case theoretical complexity is exponential.

Problema

Nombre: Linear Programming Problem
Curso: IO
Integrantes: Yo
Semestre: 2026

Mathematical Model

$$\text{máx} \quad Z = 7x1 + 3x2$$

$$\begin{aligned} 2x1 + 2x2 &\leq 9 \\ 3x1 + 1x2 &\leq 13 \\ x_j &\geq 0 \quad \text{for every variable } j \end{aligned}$$

2. Initial table

Row	Z	x_1	x_2	S_1	S_2	RHS
Z	1	-7	-3	0	0	0
R_1	0	2	2	1	0	9
R_2	0	3	1	0	1	13

Result

Optimal solution found.

3. Final Table

Row	Z	x_1	x_2	S_1	S_2	RHS
Z	1	0	0	0,5	2	30,5
R_1	0	0	1	0,75	-0,5	0,25
R_2	0	1	0	-0,25	0,5	4,25

4. Problem Solution

- An optimal solution was found.
- **Optimal Solution**
  - Objective Function Value: $Z = 30,5$.
  - `x1` $= 4,25$ (basic)
  - `x2` $= 0,25$ (basic)
  - $S_1 = 0$ (non-basic)
  - $S_2 = 0$ (non-basic)